

RACHEL N. SLAYBAUGH

DCVC ♦ 270 University Ave ♦ Palo Alto, CA 94301

rachel.slaybaugh@dcvc.com ♦ (415) · 841 · 3935

EDUCATION

Ph.D.	University of Wisconsin–Madison Nuclear Engineering and Engineering Physics, with a certificate in Energy Analysis and Policy	2011
M.S.	University of Wisconsin–Madison Nuclear Engineering and Engineering Physics	2008
B.S.	Pennsylvania State University Nuclear Engineering	2006

EXPERIENCE

Data Collective Venture Capital (DCVC) <i>Principal</i>	Feb. 2022 – present <i>Palo Alto, CA</i>
• Source, select, fund, and support climate tech companies • Independent board member: Radiant Nuclear (04/2022 - present) • Board observer: Brimstone (02/2022 - present), Fervo Energy (08/2022 - present), and Petra (08/2022)	

Lawrence Berkeley National Laboratory <i>Cyclotron Road Division Director</i>	Jan. 2021 – Dec. 2021 <i>Berkeley, CA</i>
• Source and select ~10 hard tech innovators per year for a 2-year fellowship program to turn their technology concept into a product with a positive societal impact (~\$6M/year) • Fellows embed at LBNL where we support their technical development and collaborations • Responsible for outcomes, safety, and reporting to Department of Energy and LBNL leadership	

University of California, Berkeley <i>Assistant Professor of Nuclear Engineering</i> <i>Associate Professor</i>	Berkeley, CA <i>Jan. 2014 – July 2020</i> <i>July 2020 – Dec. 2021</i>
• Founded the Nuclear Innovation Bootcamp, which brings diverse students from around the world to learn skills essential to innovation in nuclear energy • Developing numerical methods for neutral particle transport with an emphasis on supercomputing and advanced architectures; applications in reactor design, shielding, and nonproliferation • Also mentored PhDs in optimization, thermal fluids, and cryptography and anomaly detection • Published 25 journal articles, 44 refereed conference proceedings, 3 technical reports, 2 book chapters, 5 open source pieces of software, and 2 policy pieces • Graduated 9 PhD and 4 MS students; research adviser for 1 assistant project scientist, 1 postdoctoral scholar, 1 visiting scholar, and 15 undergraduate students • Won >\$2.5M as principal investigator (PI) and >\$26M as co-PI for numerical methods research • Created and support a course in which Berkeley students do hands on science experiments at under-served elementary schools in Oakland	

Advanced Research Projects Agency – Energy <i>Program Director</i>	Jan. 2017 – Oct. 2020 <i>Washington, DC</i>
• Created programs and selected teams developing technologies for advanced nuclear fission reactors, \$95M across 28 teams: MEITNER, the Nuclear OPEN+ cohort, LISE, and GEMINA	

- Provided deep technical and business guidance while managing: the nuclear teams; the TERRA and ROOTS Programs, supporting research for sensing and data analytics for above- and below-ground plant outcomes (~\$90M across 18 teams); and the FOCUS Program, supporting research for solar technologies that combine photovoltaic and concentrated solar power technologies (~\$12M across 4 teams)
- Supported program creation in other energy areas through workshops, brainstorming, and feedback

Bettis Laboratory

Senior Engineer in the Shield Design and Development group

West Mifflin, PA

Mar. 2012 – Aug. 2014

- Implemented the Forward-Weighted Consistent Adjoint Driven Importance Sampling (FW-CADIS) method and developed new resonance factor for variance reduction in Monte Carlo
- Qualified these methods and software for use in shield design to dramatically reduce time and improve accuracy in design calculations
- Scientific Software Development Committee: leader in developing internal website for sharing software carpentry tools and resources

University of Wisconsin–Madison

Research Assistant / Rickover Fellow

Madison, WI

Sept. 2006 – Nov. 2011

- Dissertation: “Acceleration Methods for Massively Parallel Deterministic Transport” where I added 3 new methods to Denovo, software from Oak Ridge National Laboratory, that are still used
- Developed two Monte Carlo source sampling methods for arbitrarily shaped plasma sources

Penn State Breazeale Reactor

Reactor Operator

University Park, PA

Aug. 2003 – Apr. 2006

- NRC licensed Reactor Operator for TRIGA Mark III reactor
- Analyzed core burn-up anomaly; calibrated gamma irradiation facilities

PROFESSIONAL SERVICE

Lab Start, Advisor, 2022-present

NREL Investor Advisory Board, 2022-present, Vice Chair 2023

National Academies of Science member of the Committee on Laying the Foundations for New and Advanced Nuclear Reactors in the United States, 2020-2022

Biden-Harris Transition Team, 2020

Good Energy Collective, Founding Board Chair, 2020-present

Activate, Leadership Council, 2021-present

UC Berkeley, Innovation & Entrepreneurship Council, 2021

Pennsylvania State University, Nuclear Alumni Advisory Council, 2020-2021

University of Michigan, NERS Department Advisory Board, 2019-2021

Nuclear Energy Advisory Committee, Appointed Member, 2016-2017

Journals, reviews, conference organizing

Nuclear Science and Engineering

Editorial Advisory Board, 2020-2022

reviewer

Annals of Nuclear Energy

reviewer

Journal of Computational Physics

reviewer

Journal of Nuclear Science and Technology	reviewer
Nuclear Engineering and Design	reviewer
Ocean Alkalinity Enhancement Engineering Award	proposal reviewer
Canadian Innovation Fund	proposal reviewer
Lawrence Livermore National Laboratory	LDRD reviewer
US DOE Consolidated Innovative Nuclear Research	proposal reviewer
US DOE Technology Commercialization Fund	proposal reviewer
Platform for Advanced Scientific Computing (PASC) Conference 21	Papers Domain Co-Chair for Engineering
ANS: Math & Comp, PHYSOR, national meetings	reviewer

American Nuclear Society, National Level

Math and Comp. Division	Chair Rotation 2016-2019; Exec. Comm. 2013-2016
Rad. Protection and Shielding Div.	Exec. Comm. 2015-2018
Young Members Group	Exec. Comm. 2014-2017
NEED Comm.	Chair 2013-2015, Vice Chair 2010-2013
Professional Divisions Comm.	Vice Chair 2012-2016
Student Sections Comm.	Chair 2010-2013, Vice Chair 2009-2010
Professional Women in ANS	Chair 2008-2010, Vice Chair 2006-2008
Board of Directors	Student Member 2007-2009
Other committee service	Membership, Bylaws and Rules, Public Info., 2013 Nominating Comm., 2014 Special Selection Comm. for Nominating Comm. Candidates

Software and Computing

The Hacker Within, UCB and UW (http://thehackerwithin.github.io/berkeley/)	Faculty Advisor 2014-2018; Co-founder 2009
Berkeley Research Computing (http://research-it.berkeley.edu/programs/berkeley-research-computing)	User Advisory Group 2016-present
Software and Data Carpentry (http://software-carpentry.org/)	Instructor since 2013
Berkeley Computing and Computer Science Education Committee	2014-2021

Energy and Science

Society of Industrial and Applied Mathematics	Member 2009-2020
Berkeley Energy and Resources Collaborative (BERC)	Advisory Board Member 2017-2021
UCB-ANS	Faculty Advisor 2014-2019
Nuclear Engineering Student Delegation	Co-Vice Chair 2010, Selected 2009
UW-Energy Hub	Conference Speaker Chair 2009, Founding Member 2007, Collegiate Energy Association 2008-2010

Education

UC Berkeley Admissions Reader	2021
UCB Designated Emphasis in Energy Science and Technology Executive Committee	2020-2021
Berkeley College of Engineering Graduate Education Committee	2014-2017
Berkeley College of Engineering Undergraduate Education Committee	2020-present

TEACHING EXPERIENCE

University of California, Berkeley

Jan. 2014 - Present

*Associate Professor of Nuclear Engineering**Berkeley, CA*

- Founder Nuclear Innovation Bootcamp, Su. 2016 - 2022
- NE 250, Nuclear Reactor Theory (graduate-level): Fa. 2015, 2017
- NE 255, Numerical Simulations for Radiation Transport (graduate-level): Fa. 2016
- NE 155, Introduction to Numerical Simulations for Radiation Transport (senior-level elective): Sp. 2014, 2015, 2016, 2017, 2021
- NE 100, Introduction to Nuclear Energy and Technology (junior-level required): Fa. 2020
- NE 198, Faculty sponsor for class in which Berkeley students do hands-on science experiments with students in under-served elementary schools in Oakland: Fa. 2015 - Fa. 2021
- NE 24, Putting the Science in Computational Science, Sp. 2015, 2016, 2017

Software Carpentry Scientific Computing Workshops*Instructor*

- Jan. 14-15, 2016: git; Berkeley Institute for Data Science
- July 16, 2015: shell; École Polytechnique Fédérale Lausanne
- July 1-2, 2015: shell and Python; for underrepresented minority students; UC, Berkeley
- June 11, 2015: Python; Oak Ridge National Laboratory
- Jan. 5-6, 2015: version control; for women only; University of Colorado, Boulder
- Apr. 14-15, 2014: introductory material, version control, object oriented concepts in Python; for women only; Lawrence Berkeley National Laboratory

Bettis Laboratory

Mar. 2012 - Aug. 2014

*Instructor**West Mifflin, PA*

- Qualified instructor for Bettis Reactor Engineering School (BRES), an internal school for new DOE-Naval Reactors employees
- Co-taught BRES Shielding course Fa. 2012, 2013, and Sp. 2013

University of Pittsburgh

Fa. 2012, Sp. 2013

*Adjunct Professor**Pittsburgh, PA*

- Co-taught Introduction to Nuclear Engineering (ENGR 1700), Fa. 2012
- Co-taught *new* course Nuclear Chemistry and Radiochemistry (ENGR 2112), Sp. 2013

SELECTED PRESENTATIONS

Guest Lecture “Nuclear Energy.” for Energy 790-1 Emerging Energy Technologies Class at Duke University. Online. 22 Sept. 2022. (invited)

“Venture Capital for Climate Tech.” Colorado School of Mines NE Program Graduate Colloquium. Online. 20 Sept. 2022. (invited)

Matt Trevithick and Rachel Slaybaugh “The future of nuclear energy from a venture capital perspective.” Denmark Bridge CEO Summit. Sausalito, CA. 10 May 2022. (invited)

Guest Lecture “The Debate on Nuclear Power.” for Security Policy Course in the Goldman School of Public Policy. 17 Nov. 2021. (invited)

Guest Lecture “Nuclear Energy (where we are, where we are going, and why)” for E93 Class at UC Berkeley. Online. 1 Oct. 2021. (invited)

“Cyclotron Road.” Office of Technology Transitions MACOP Program. Online. 16 Sept. 2021. (invited)

“How Can Nuclear Energy Help Fight Climate Change?” American Institute of Chemical Engineers (AIChE) Virtual Section. Online. 25 Aug. 2021. (invited)

Guest Lecture “Nuclear Energy.” for The Institute for Security and Conflict Studies (ISCS) fellowship program at George Washington University. Online. 16 July 2021. (invited)

“Optimal Operations of Advanced Reactors: How Do We Get There?” Current Issues in Computational Methods – Roundtable, American Nuclear Society Annual Meeting. Online. 14 June 2021. (invited)

“Cyclotron Road.” The Oppenheimer Science and Energy Leadership Program. Online. 19 Apr. 2021. (invited)

Panelist for “Serving the Nation: Engineering as Leaders in Government.” Spring Dean’s Society Event. Online 24 Mar. 2021. (invited)

“How Can Nuclear Support a Clean Energy Future?” American Physical Society Meeting. Online. 17 Mar. 2021. (invited)

“Nuclear in the 2020s: considerations for the decade ahead.” **Keynote Speaker**, ARPA-E Annual Fission Meeting. Online. 23 Feb. 2021. (invited)

Guest Lecture “Nuclear Energy (where we are, where we are going, and why)” for STIA 402: Clean Energy Innovation, Georgetown University School of Foreign Service. Online. 20 Oct. 2020. (invited)

Panelist for “Flexible Nuclear for Future Energy Systems Webinar.” Lucid Catalyst and ClearPath. 5 Aug. 2020. (invited)

Panelist for “Beyond Now: Microreactors and Near Term Demonstrations - Perspectives from the Department of Defense and Department of Energy.” Panelist. U. S. WIN Virtual Conference. 28 July 2020. (invited)

Panelist for “Advanced Nuclear Tech in a Decarbonized Energy Future: Feasible or Fantastical?” Duke Energy Initiative lecture series. Online. 14 July 2020. (invited)

Panelist for “Challenges and Opportunities in Thermal Hydraulics of High Temperature Gas Cooled Reactors.” American Nuclear Society Annual Meeting. Online. 8 June 2020. (invited)

Panelist for “Building Your Leadership Competency.” American Nuclear Society Annual Meeting. Online. 8 June 2020. (invited)

“How Can Nuclear Be Part of a Clean Energy Future?” G20 Workshop on Circular Carbon Economy. Riyadh, KSA. 5 March 2020. (invited)

“Special Update: ARPA-E Nuclear Program Update.” Advanced Reactors Summit VII. Knoxville, TN. 11 Feb 2020. (invited)

“The Opportunity of Next-Gen Nuclear Energy.” Citizen’s Climate Lobby, Fairfax Meeting. Annapolis, VA. 11 Feb 2020. (invited)

“How Can Nuclear Energy Be Part of a Clean Energy Future?” Climate Tech Convening 2020. San Francisco, CA. 30 Jan 2020. (invited)

Panelist for “Time to be Inclusive and Pragmatic Before It’s Too Late: Advanced nuclear technologies must be a significant part of the solution set.” Cleantech Forum. San Francisco, CA. 28 Jan 2020. (invited)

“What is ARPA-E Doing in Nuclear Energy?” Argonne National Laboratory Colloquium. Lemont, IL. 20 November 2019. (invited)

Guest Lecture “Nuclear Energy” for STIA 402: Clean Energy Innovation, Georgetown University School of Foreign Service. 23 Oct. 2019. (invited)

Panelist for “The role of innovation in the civilian and military nuclear energy sectors.” The civilian-military nexus and the role of innovation in nuclear energy, Atlantic Council. Washington, DC. 15 Oct 2019. (invited)

“Nuclear and ARPA-E: Activities and Opportunities.” NUC Workshop: Innovations in Advanced Reactor Design, Analysis, and Licensing. Raleigh, NC (given remotely). 18 September 2019. (invited)

“Nuclear Energy Programs and Research at Advanced Research Projects Agency-Energy (ARPA-E).” GABI nuclear energy roundtable. Washington, DC. 12 September 2019. (invited)

Panelist for “Future Scientists and Engineers for Nuclear Energy.” Third Symposium on US-Japan Nuclear Energy Research Cooperation. Washington, DC. 26 June 2019. (invited)

Panelist for “Lessons from Solar PV.” Breakthrough Dialogue 2019 Whole Earth Discipline. Sausalito, CA. 21 June 2019. (invited)

“Computing in Nuclear Engineering: Moving Beyond Today’s Limitations.” MIT Nuclear Science & Engineering Colloquium. Cambridge, MA. 29 April 2019.

“Creating the Future of Nuclear Energy.” CITRIS Research Exchange Seminar Series. Berkeley, CA. 28 Nov. 2018. (invited)

“U.S. Nuclear Innovation System.” Commercialisation of Small Nuclear in the UK. Coventry, UK. 6 Nov. 2018. (invited)

“MEITNER: The Newest ARPA-E Program.” NASA iTech. New York, NY. 14 June 2018. (invited)

“Deep decarbonization: financing the rapid deployment of innovations after R&D.” 2050 Today Panel. San Francisco, CA. 15 June 2018. (invited)

“Advanced Solvers for Radiation Transport.” University of Illinois Urbana Champaign NPPE Dept. Graduate Colloquium. Champaign, IL. 5 Dec. 2017. (invited)

“Nuclear Energy: What Can Analytics Do for Economics and Safety?” **Keynote Speaker**, Industrial Machine Learning Workshop 17. San Francisco, CA. 24 Oct. 2017. (invited)

Panelist for “Alternative energy, nuclear and CCS: What are the prospects?” Low-Emissions Solution Conference at the 5th International Conference on Sustainable Development. New York, NY. 21 Sept. 2017. (invited)

Panelist for “The Case for Nuclear: Communications and Analyzing Risks and Benefits” Aspen Institute Future of Nuclear Energy Roundtable. Aspen, CO. 12 Sept. 2017. (invited)

“Advanced Solvers and Innovation for Penetrating Radiation” **Keynote Speaker**, SPIE Penetrating Radiation Technical Event. San Diego, CA. 8 Aug. 2017. (invited)

“Challenges & Opportunities in Developing an Innovation Culture in Nuclear.” Second Annual Third Way Advanced Nuclear Summit Panel. Washington, DC. 21 Feb. 2017. (invited)

“Advancing Reactors with Advanced Solvers and Innovative Programs.” Electric Power Research Institute. Research Meeting. Palo Alto, CA. 13 Feb. 2017. (invited)

“Exploring the Advanced Nuclear Ecosystem.” SxSW Eco Panel. Austin, TX. 12 Oct. 2016. (invited)

- “The Opportunity of Global Nuclear Innovation.” 49th Japan Atomic Industrial Forum (JAIF) Annual Conference. Tokyo, Japan. 12 April 2016. (invited)
- “Exascaling Nuclear Innovation.” **Keynote speaker**, Conference on Data Analysis 2016. Santa Fe, NM. 4 Mar. 2016. (invited)
- “Let’s Get Real: When can we expect commercial advanced reactors?.” The Advanced Nuclear Summit and Showcase Panel. Washington, DC. 27 Jan. 2016. (invited)
- “Hybrid Transport Methods for Shielding Challenges.” Idaho National Laboratory seminar. Idaho Falls, ID. 20 Nov. 2015. (invited)
- K.L. Rowland, R.N. Slaybaugh. “Developments in the GPU-accelerated WARP Monte Carlo Neutron Transport Code.” Workshop sur l’utilisation des accélérateurs (GPUs, MICs) pour les simulations Monte-Carlo. École Polytechnique, Saclay, France. 10 July 2015.
- “The PyNE Software Library: A Framework for ENSDF?” Nuclear Data Week Meeting. Brookhaven National Laboratory. 6 Nov. 2014.
- “The Resonance Factor Method: Accelerating Monte Carlo in the Presence of Space and Energy Self-Shielding.” CEA-Saclay Colloquium. Saclay, France. 26 June 2014.
- R.N. Slaybaugh, T.M. Evans, P.P.H. Wilson, S.C. Wilson. “Radiation Transport: Computational Methods and Real-World Use.” NC State Univ. NE Dept. Graduate Colloquium. Raleigh, NC. 8 Nov. 2012. (invited)
- R. Slaybaugh, M. Arbidze, S. Lamichhane, D. O’Connor. “An Evaluation of European Union Energy Policies.” UW–Madison Center for World Affairs and the Global Economy Seminar. Madison, WI. 11 May 2011.
- R.N. Slaybaugh, T.M. Evans, G.G. Davidson. “Parallel Algorithms for Fixed-Source and Eigenvalue Problems.” 2010 SIAM Annual Meeting. Pittsburgh, PA. 12-16 July 2010.

COMPUTER SKILLS

Languages	C++, Python, Fortran 90/95/2003
Versioning and Testing	git, svn, cvs, CTest, GoogleTest, nose, TravisCI
Tools	EC2, Doxygen, L ^A T _E X, MathCAD, Mathematica, shell, vim, bash, Emacs, Jupyter, Trilinos, LAPACK, MPI, Valgrind, cmake
Nuclear Software	Exnihilo, ADVANTG, MCNP, Serpent, SCALE, PyNE, MOOSE

HONORS AND AWARDS

American Nuclear Society (ANS) Young Member Excellence Award	2014
ANS Presidential Citation	2014
Rickover Fellowship	2008-2011
Second Place, 2011 ANS Winter Meeting Poster Session	2011
Selected participant, Modeling Experimentation and Validation Reactor Physics School	Jul. 2011
Selected participant, Energy Hub conference Poster Session	2011
Everitt P. Blizzard Memorial Scholarship, ANS	2010-2011
ANS Mathematics and Computation Division Best Summary + Presentation Award	Nov. 2010
Graduate Scholarship, ANS	2009-2010
Selected participant, Lindau Meeting of Nobel Laureates in Physics	2008

PUBLICATIONS

Italicized names indicate my students or researchers

- A. *J. Novak*, Sebastian Schunert, Robert Carlsen, Paolo Balestra, Rachel Slaybaugh, Richard Martineau. "Multiscale Thermal-Hydraulic Modeling of the Pebble Bed Fluoride-Salt-Cooled High-Temperature Reactor." *Annals of Nuclear Energy*. **154** (2021) 107968.
<https://www.sciencedirect.com/science/article/pii/S0306454920306642>
- A. *J. Novak*, R. W. Carlsen, S. Schunert, P. Balestra, R. N. Slaybaugh, and R. C. Martineau. "Pronghorn: A Multidimensional Coarse Mesh Application for Advanced Reactor Thermal-Hydraulics." *Nuclear Technology*. DOI: 10.1080/00295450.2020.1825307
<https://www.tandfonline.com/doi/full/10.1080/00295450.2020.1825307>
- R. Martineau, D. Andrs, R. Carlsen, D. Gaston, J. Hansel, F. Kong, A. Lindsay, C. Permann, A. Slaughter, E. Merzari, R. Hu, A. *Novak*, R. Slaybaugh. "Multiphysics for Nuclear Energy Applications Using a Cohesive Computational Framework." *Nuclear Engineering and Design*. **367** (2020) 1107512.
<https://doi.org/10.1016/j.nucengdes.2020.110751>
Best research award from ScienceDirect
- A. *J. Novak*, J. W. Peterson, L. Zou, D. Andrš, R. N. Slaybaugh, R. C. Martineau, "Validation of Pronghorn Friction-Dominated Porous Media Thermal-Hydraulics Model with the SANA Experiments." *Nuclear Engineering and Design*. **350** (2019) 182-194.
<https://www.sciencedirect.com/science/article/pii/S0029549319301037>
- Nicholas J. Quartemont, *James E. Bevins*, Lee Bernstein, Rachel Slaybaugh. "Analysis of an Energy Tuning Assembly for Simulating Nuclear Weapon Environments at the National Ignition Facility." *Journal of Radiation Effects*. **38**:1 (2020) 13-26.
<https://apps.dtic.mil/sti/pdfs/AD1093191.pdf>
- Marissa Ramirez de Chanlette, *Weixiong Zheng*, R. N. Slaybaugh. "A Two-Grid and Nonlinear Diffusion Acceleration Method for the SN Equations with Neutron Upscattering." *Journal of Computational Transport Theory*. **49**:1 (2020) 1-14.
<https://doi.org/10.1080/23324309.2019.1695631>
- Mario Ortega, Rachel N Slaybaugh, Peter N Brown, Teresa S Bailey, Britton Chang. "A Rayleigh Quotient Method for Criticality Eigenvalue Problems in Neutron Transport." *Annals of Nuclear Energy*. **138** (2020) 107120.
<https://doi.org/10.1016/j.anucene.2019.107120>
- A. *J. Novak*, J. W. Peterson, L. Zou, D. Andrš, R. N. Slaybaugh, R. C. Martineau, "Validation of Pronghorn Friction-Dominated Porous Media Thermal-Hydraulics Model with the SANA Experiments." *Nuclear Engineering and Design*. **350** (2019) 182-194.
<https://www.sciencedirect.com/science/article/pii/S0029549319301037>
- Richard Vasques, Leonardo R. C Moraes, Ricardo C Barros, Rachel N Slaybaugh, "A Spectral Approach for Solving the Nonclassical Transport Equation." *Journal Of Computational Physics*. **402** (2020) 109078.
<http://arxiv.org/abs/1812.04811>
<https://www.sciencedirect.com/science/article/pii/S0021999119307831>
- Madicken Munk, Rachel Slaybaugh, "Review of Hybrid Methods for Deep-Penetration Neutron Transport." *Nuclear Science and Engineering*. **193**:10 (2019) 1055-1089.
<https://www.tandfonline.com/doi/full/10.1080/00295639.2019.1586273>
- J. S. *Rehak*, L. M. Kerby, M. D. DeHart, R. N. Slaybaugh. "Weighted Delta-Tracking with Scattering," *Nuclear Engineering and Design*. **342** (2019) 231-239.

<https://arxiv.org/abs/1802.02237>

- James Bevins, Zachary Sweger, Ninad Munshi, Bethany Goldblum, Josh Brown, Darren Bleuel, Lee Bernstein, Rachel Slaybaugh. "Performance Evaluation of an Energy Tuning Assembly for Neutron Spectral Shaping." *Nuclear Inst. and Methods in Physics Research, A*. **923** (2019) 79-87.
<https://www.sciencedirect.com/science/article/pii/S0168900219300968>
- Kelly L. Rowland, Cory D. Ahrens, Steven Hamilton, and R.N. Slaybaugh. "Assessment of the Lagrange Discrete Ordinates Equations for Three-Dimensional Neutron Transport" *Nuclear Science and Engineering*. **193**:3 (2019) 233-252.
<https://github.com/kellyrowland/lao-deterministic>
- James E. Bevins, R.N. Slaybaugh. "Gnowee: A Metaheuristic Optimization Algorithm for Solving Engineering Problems Containing Continuous and Discrete Design Parameters." *Nuclear Technology*. **205**:4 (2019) 542-562.
<http://arxiv.org/abs/1804.05429>
- I. Makine, R. Vasques, R.N. Slaybaugh. "Exact Transport Representation of the Classical and Nonclassical Simplified P_N Equations." *Journal of Computational and Theoretical Transport*. **47**:4-6 (2018) 326-349.
<https://www.tandfonline.com/doi/abs/10.1080/23324309.2018.1496938>
- R.N. Slaybaugh, M. Ramirez-Zweiger, Tara Pandya, Steven Hamilton, T.M. Evans. "Eigenvalue Solvers for Modeling Nuclear Reactors on Leadership Class Machines," *Nuclear Science and Engineering*. **190** (2017) 31-44.
<https://arxiv.org/abs/1708.04928>
- Jeffery B. Greenblatt, Nicholas R. Brown, Rachel Slaybaugh, Theresa Wilks, Emma Stewart, and Sean T. McCoy. "The Future of Low-Carbon Electricity," *Annual Review of Environment and Resources*. **42** (2017) 289-316.
<http://www.annualreviews.org/doi/10.1146/annurev-environ-102016-061138>
- Ryan M. Bergmann, Kelly L. Rowland, Nikola Radnović, Rachel N. Slaybaugh, Jasmina L. Vujić. "Performance and Accuracy of Criticality Calculations Performed Using WARP, A Framework for Continuous Energy Monte Carlo Neutron Transport in General 3D Geometries on GPUs," *Annals of Nuclear Energy*. **103** (2017) 334-349.
<https://www.sciencedirect.com/science/article/pii/S0306454916309902>
- Leah E. Morgan, Madicken Munk, Brett Davidheiser-Kroll, Nicholas H. Warner, Sanjeev Gupta, Rachel Slaybaugh, Patrick Harkness, Darren F. Mark. "Instrumentation development for planetary in situ $^{40}\text{Ar}/^{39}\text{Ar}$ geochronology," *Geostandards and Geoanalytical Research*. **41**:3 (2017) 381-396.
<https://onlinelibrary.wiley.com/doi/full/10.1111/ggr.12170>
- R. Vasques, K. Krycki, R. N. Slaybaugh. "Nonclassical Particle Transport in One-Dimensional Random Periodic Media," *Nuclear Science and Engineering*. **185**:1 (2017) 78-106.
<https://arxiv.org/abs/1602.00825>
- S.C. Wilson and R.N. Slaybaugh. "Improved Monte Carlo Variance Reduction for Space and Energy Self-Shielding," *Nuclear Science and Engineering*. **179**:1 (2015) 22-41.
<https://arxiv.org/abs/1502.04749>
- G.G. Davidson, T.M. Evans, J.J. Jarrell, S.P. Hamilton, T.M. Pandya, and R.N. Slaybaugh, "Massively Parallel, Three-Dimensional Transport Solutions for the k-Eigenvalue Problem," *Nuclear Science and Engineering*. **177**:2 (2014) 111-125.
<https://www.tandfonline.com/doi/abs/10.13182/NSE12-101>

- R.N. Slaybaugh, T.M. Evans, G.G. Davidson, and P.P.H. Wilson. “Multigrid in energy preconditioner for Krylov solvers,” *Journal of Computational Physics*. **242** (2013) 405-419.
<https://arxiv.org/abs/1612.00907>
- T.M. Evans, A.S. Stafford, R.N. Slaybaugh, and K.T. Clarno. “Denovo—A new three-dimensional parallel discrete ordinates code in SCALE.” *Nuc. Tech.* **171**:2 (2010) 171-200.
<https://www.tandfonline.com/doi/abs/10.13182/NT171-171>
- R.N. Slaybaugh, P.P.H. Wilson, L.A. El-Guebaly, and E.P. Marriott. “Three-Dimensional Neutron Source Models for Toroidal Fusion Energy Systems.” *Fusion Engineering and Design*. **84** (2009) 1774-1778.
<https://www.sciencedirect.com/science/article/pii/S0920379609000076>

Refereed Conference Proceedings

- Grey Batie, *Vanessa Goss*, Cameron Cage, Christopher Poresky, Per Peterson, Rachel Slaybaugh, Kai Vetter. “Determining the Optimal Radiation Detector Configuration for Measuring Fuel Characteristics of an Accountability Tank within a PUREX Facility.” Proceedings of Institute of Nuclear Materials Management 63rd Annual Meeting, virtual meeting, July 2022.
- Grey Batie, *Vanessa Goss*, Christopher Poresky, Per Peterson, Rachel Slaybaugh, Kai Vetter. “Detecting and Quantifying Diversion in an Experimental Aqueous Reprocessing Loop.” Proceedings of Institute of Nuclear Materials Management 63rd Annual Meeting, virtual meeting, July 2022.
- Mitch Negus*, David Farley, Rachel Slaybaugh. “No Data, No Problem. Giving nuclear inspectors better tools without revealing state secrets.” Proceedings of ENIGMA, A Usenix Conference in Oakland, CA, Feb 2021.
<https://www.usenix.org/conference/enigma2021/presentation/negus>
- Grey Batie, *Vanessa Goss*, Laura Shi, Christopher Poresky, Lucian Mihalescu, Per Peterson, Rachel Slaybaugh, and Kai Vetter. “Fault Detection Methods To Accurately Detect And Quantify Holdup In Advanced Nuclear Material Recycle Facilities.” Proceedings of Institute of Nuclear Materials Management 61st Annual Meeting in Baltimore, MD, July 2020.
- Mitch Negus*, Rachel Slaybaugh, and David Farley. “Garbled circuits for enabling privacy preserving safeguards.” Proceedings of Institute of Nuclear Materials Management 61st Annual Meeting in Baltimore, MD, July 2020.
- J. S. Rehak*, R. N. Slaybaugh. “Assessing the Effectiveness of Acceleration Methods for Deterministic Neutron Transport Solvers.” Proceedings of the 2020 ANS Annual Meeting in Phoenix, AZ (online), June 2020. Transactions vol. 122.
- Kelly L. Rowland*, Cory D. Ahrens, Steven Hamilton, and R.N. Slaybaugh. “Assessment of the Lagrange Discrete Ordinates Equations for Monte Carlo Variance Reduction Parameter Generation.” Proceedings of Joint International Conference on Supercomputing in Nuclear Applications + Monte Carlo 2020 in Tokyo, Japan, May 2020.
- N. J. Quartemont, *J. E. Bevins*, R. Slaybaugh, and L. Bernstein, “Analysis of an Energy Tuning Assembly for Simulating Nuclear Weapon Environments at the National Ignition Facility.” in Proceedings of the Hardened Electronics and Radiation Technology Conference in Melbourne, FL, Nov 2019. [winner Best Student Paper Award]
- A.J. Novak*, R.N. Slaybaugh, and R.C. Martineau. “Multiscale Core Thermal-Hydraulics Analysis of the Pebble Bed Fluoride-Salt-Cooled High-Temperature Reactor (PB-FHR).” Proceedings of the The International Conference on Mathematics and Computational Methods applied to Nuclear Science and Engineering in Portland, OR, Aug 2019.
- R. Martineau, D. Andrs, R. Carlsen, D. Gaston, J. Hansel, F. Kong, C. Permann, E. Mezari, Rui

- Hu, A. Novak, R. Slaybaugh. “Multiphysics for Nuclear Energy Applications Using a Cohesive Computational Framework.” 18th International Topical Meeting on Nuclear Reactor Thermal Hydraulics – NURETH in Portland, OR, Aug 18-23, 2019.
- April Novak, Josh Peterson, Ling Zou, Rachel Slaybaugh, Rich Martineau. “Porous Media Thermal Hydraulics Simulations of Pebble Bed Nuclear Reactors using Pronghorn.” SIAM Conference on Computational Science and Engineering in Spokane, WA, Feb 25 – Mar 1, 2019. [invited]
https://meetings.siam.org/sess/dsp_programsess.cfm?SESSIONCODE=66138
- Kelly L. Rowland, Cory Ahrens, Steven Hamilton, Rachel Slaybaugh. “Assessment of the Lagrange Discrete Ordinates Equations for Three-dimensional Neutral Particle Transport.” 2019 SIAM Conference on Computational Science and Engineering in Spokane, WA, Feb 25 – Mar 1, 2019. [invited]
https://meetings.siam.org/sess/dsp_talk.cfm?p=96180
- Nicholas J. Quartemont, James E. Bevins, Rachel Slaybaugh, Lee Bernstein. “Development of a Novel National Ignition Facility Platform for Simulating Nuclear Relevant Neutron Environments.” IEEE Nuclear Science Symposium and Medical Imaging Conference in Sydney, Australia, Nov 2018.
- April J. Novak, Ling Zou, John W. Peterson, Richard C. Martineau, and Rachel N. Slaybaugh. “Pronghorn: Porous Media Thermal-Hydraulics for Reactor Applications.” Proceedings of the 2018 ANS Winter Meeting in Orlando, FL, Nov 2018. Transactions vol. 119. [invited]
- M. I. Ortega, P. N. Brown, T. S. Bailey, and B. Chang, and R. N. Slaybaugh. “A Rayleigh Quotient Method for Criticality Eigenvalue Problems in Neutron Transport.” Proceedings of PHYTRA4 - The Fourth International Conference on Physics and Technology of Reactors and Applications in Marrakech, Morocco, Sept 17-19, 2018. [invited]
- Sandra Bogetic, James E. Bevins, Lee A. Bernstein, Rachel Slaybaugh, and Jasmina Vujić. “Meta-heuristic Optimization Method for Neutron Spectra Shaping.” Proceedings of the 2018 ANS June Meeting in Philadelphia, PA, June 2018. Transactions vol. 118.
- A. J. Novak, L. Zou, J. W. Peterson, R. C. Martineau, and R. N. Slaybaugh. “Pronghorn: A Porous Media Thermal-Hydraulics Core Simulator and its Validation with the SANA Experiments.” Proceedings of the International Congress on Advances in Nuclear Power Plants in Charlotte, NC, Apr 2018.
- A. Novak, P. Romano, B. Wendt, R. Rahaman, E. Merzari, L. Kerby, C. Permann, R. Martineau, and R. N. Slaybaugh. “Preliminary Coupling of OpenMC and Nek5000 Within The MOOSE Framework.” Proceedings of the PHYSOR 2018 Meeting in Cancun, Mexico, Apr 2018.
- I. Makine, R. Vasques, and R.N. Slaybaugh. “Exact Transport Representations of the Classical and Nonclassical Simplified P_N Equations.” 25th International Conference on Transport Theory, Monterey, CA, 16-20 Oct 2017.
- M. I. Ortega, P.N. Brown, T. S. Bailey, R. N. Slaybaugh, and B. Chang. “A Rayleigh Quotient Method for Solving the Alpha-Eigenvalue Problem in Neutron Transport.” 25th International Conference on Transport Theory, Monterey, CA, 16-20 Oct 2017.
- Marissa Ramirez Zweiger, Weixiong Zheng, and R.N. Slaybaugh. “Two-Grid and Nonlinear Diffusion Acceleration Method for the Multigroup S_N Equations with Neutron Upscattering.” 25th International Conference on Transport Theory, Monterey, CA, 16-20 Oct 2017.
- M. Wrenninge, R. Vasques, R.N. Slaybaugh. “A Generalized Volume Rendering Approach for Computer Graphics.” 25th International Conference on Transport Theory, Monterey, CA, 16-20 Oct 2017.

- J.S. Rehak*, L.M. Kerby, M.D. DeHart, R.N. Slaybaugh, J. Leppänen. “Implementation of Weighted Delta-Tracking with Scattering in the Serpent 2 Monte Carlo Code.” Proceedings of the 2017 ANS June Meeting in San Francisco, CA, June 2017. Transactions vol. 116.
- Weixiong Zheng*, Ryan McClarren, Rachel Slaybaugh. “A Continuous-Discontinuous Hybrid Finite Element Method for Solving Radiation Transport.” Proceedings of the 2017 ANS June Meeting in San Francisco, CA, June 2017. Transactions vol. 116.
- Kelly L. Rowland*, *Ryan M. Bergmann*, Rachel N. Slaybaugh, Jasmina L. Vujić. “Delta-tracking in the GPU-accelerated WARP Monte Carlo Neutron Transport Code.” International Conference on Mathematics & Computational Methods Applied to Nuclear Science & Engineering, Jeju, South Korea, Apr 2017. [invited]
- R. Vasques* and R. N. Slaybaugh. “Simplified P_N Equations For Nonclassical Transport With Isotropic Scattering.” International Conference on Mathematics & Computational Methods Applied to Nuclear Science & Engineering, Jeju, South Korea, Apr 2017. [invited]
<https://arxiv.org/abs/1610.04314>
- Richard Vasques*, Rachel Slaybaugh, Kai Krycki, “Nonclassical Particle Transport in the 1-D Diffusive Limit.” Proceedings of the 2016 ANS June Meeting in New Orleans, LA, June 2016. Transactions vol. 114.
<https://arxiv.org/abs/1601.02495>
- M. Munk*, R.N. Slaybaugh, Tara M. Pandya, Seth R. Johnson, T. M. Evans, “FW/CADIS- Ω : An Angle-Informed Hybrid Method for Deep-Penetration Radiation Transport.” Proceedings of the PHYSOR 2016 Meeting in Sun Valley, ID, May 2016.
<https://arxiv.org/abs/1612.00793>
- J. Bevins*, R. Slaybaugh, L. Bernstein, E. Henry, W. Dunlop, “Targeted Modification of Neutron Energy Spectra for National Security Applications.” Proceedings of the 2016 Hardened Electronics And Radiation Technology Technical Interchange Meeting in Monterey, CA, Apr 2016.
- J. Bevins*, R. Slaybaugh, L. Bernstein, W. Dunlop, E. Henry. “Application of Metaheuristic Optimization Methods for Neutron Spectral Shaping Applications.” Proceedings of the Conference on Data Analysis 2016 in Santa Fe, NM, Mar 2016.
- K. L. Rowland*, R. N. Slaybaugh, *R. M. Bergmann*, and J. Vujić, “Implementing delta-tracking in a GPU-accelerated Monte Carlo neutron transport,” Proceedings of Frontiers in Computational Physics: Energy Sciences in Zurich, Switzerland, June 2015.
- R.N. Slaybaugh, T.M. Evans, G.G. Davidson, and P.P.H. Wilson, “Rayleigh Quotient Iteration with a Multigrid in Energy Preconditioner for Massively Parallel Neutron Transport,” Proceedings of Joint International Conference on Mathematics and Computation, Supercomputing in Nuclear Applications, and the Monte Carlo Method in Nashville, TN, Apr 2015.
<https://arxiv.org/abs/1702.02111>
- M. Munk*, L. Morgan, R. Slaybaugh, B. Davidheiser-Kroll, K. van Bibber, and D. Mark, “Design and Feasibility Study of a Compact Neutron Source for Extraterrestrial Geochronology Applications,” Proceedings of Joint International Conference on Mathematics and Computation, Supercomputing in Nuclear Applications, and the Monte Carlo Method in Nashville, TN, Apr 2015.
- Elliott Biondo, Anthony Scopatz, Matthew Gidden, Rachel Slaybaugh, and Cameron Bates. “Quality Assurance within the PyNE Open Source Toolkit,” Proceedings of the 2014 ANS Winter Meeting in Anaheim, CA, Nov 2014. Transactions vol. 111.
- Cameron Bates, Elliott Biondo, Kathryn Huff, and et al. “PyNE Progress Report,” Proceedings of the 2014 ANS Winter Meeting in Anaheim, CA, Nov 2014. Transactions vol. 111.

- S.C. Wilson and R.N. Slaybaugh. “Monte Carlo Importances in the Presence of Space and Energy Self-Shielding,” Proceedings of the 2013 ANS Winter Meeting in Washington, DC, Nov 2013. Transactions vol. 109.
- R.N. Slaybaugh and S.C. Wilson. “Deterministic Parameter Study for Fixed-Source Calculations Using FW-CADIS,” Proceedings of the 2013 ANS Annual Meeting in Atlanta, GA, June 2013. Transactions vol. 108.
- R.N. Slaybaugh, T.M. Evans, G.G. Davidson, and P.P.H. Wilson. “Rayleigh Quotient Iteration in 3D, Deterministic Neutron Transport,” Proceedings of the PHYSOR 2012 Meeting in Knoxville, TN, Apr 2012.
- P.J. Snouffer, R.N. Slaybaugh, and P.P.H. Wilson. “Criticality Benchmark Comparisons for DAGMC,” Proceedings of the 2011 ANS Annual Meeting in Hollywood, FL, June 2011. Transactions vol. 104.
- G.G. Davidson, T.M. Evans, J.J. Jarrell, and R.N. Slaybaugh, “Massively Parallel, Three-Dimensional Transport Solutions for the k-Eigenvalue Problem,” Proceedings of the International Conferences on Mathematics and Computational Methods Applied to Nuclear Science and Engineering in Rio de Janeiro, RJ, Brazil, May 2011.
- G.G. Davidson, T.M. Evans, R.N. Slaybaugh, and C.G. Baker. “Massively Parallel Solutions to the k-Eigenvalue Problem,” Proceedings of the 2010 ANS Winter Meeting in Las Vegas, NV, Nov 2010. Transactions vol. 103. [winner of Mathematics and Computation Division “Best Summary + Presentation” award]
- T.M. Evans, G.G. Davidson, and R.N. Slaybaugh. “Three-Dimensional Full Core Power Calculations for Pressurized Water Reactors,” Proceedings of the 2010 Scientific Discovery through Advanced Computing (SciDAC) Conference. Chattanooga, TN, 11-15 July, 2010. Oak Ridge National Laboratory.
- R.N. Slaybaugh, “Scouting Programs for Educational Outreach,” Proceedings of the 2009 ANS Winter Meeting in Washington, DC, Nov 2009. Transactions vol. 101.
- R.N. Slaybaugh, E.P. Marriott, P.P.H. Wilson, and L. El-Guebal, “A Study of the Effects of Source Sampling Methods on ARIES-RS NWL Profiles,” Proceedings of the ARIES-Pathways Project Meeting, 28-29 May 2008, Madison WI.
- R.N. Slaybaugh, M.L. Williams, D. Ilas, D.E. Peplow, B.L. Kirk, T.L. Nichols, Y.Y. Azmy, and M.P. Langer, “Radiation Treatment Planning Using Discrete Ordinates Codes,” Proceedings of the 2007 ANS Annual Meeting in Boston, MA, June 2007. Transactions vol. 96.
- R. Slaybaugh. “Strengths and Weaknesses of Nuclear Engineering Education,” presented at 2007 ANS Annual Meeting in Boston, MA, June 2007. Transactions vol. 96.

Technical Reports

- David R. Farley, *Mitch G. Negus*, Rachel N. Slaybaugh. “Industrial Internet-of-Things & Data Analytics for Nuclear Power & Safeguards.” Sandia National Laboratories, SAND2018-12807, November 2018.
- A.J. Novak, L. Zou, J.W. Peterson, D. Andrs, J. Kelly, R.N. Slaybaugh, R.C. Martineau, and H.D. Gougar. “Pronghorn Theory Manual.” Idaho National Laboratory, INL/EXT-18-44453, January 2018.
- L. Bernstein, D. Brown, et al. “Nuclear Data Needs and Capabilities for Applications.” White Paper. Lawrence Berkeley National Laboratory, May 27-29 2015.
<https://arxiv.org/abs/1511.07772>

Book Chapters

Slaybaugh, Rachel; Lovering, Jessica; Baker, Suzanne. “How Innovative New Reactors Could Improve Public Acceptance” in *Encyclopedia of Nuclear Energy* (Vol. 4 pg. 682-691), edited by Andrew Kadak and Ehud Greenspan, Elsevier, 2021.

<https://www.sciencedirect.com/science/article/pii/B9780128197257000842>

Slaybaugh, Rachel. “Reproducible Computational Science on High Performance Computers.” *The Practice of Reproducible Research, Case Studies and Lessons from the Data-Intensive Sciences*, edited by Justin Kitzes, Daniel Turek, and Fatma Deniz, UC Press, 2017.

<https://www.practicereproducibleresearch.org/case-studies/slaybaugh.html>

Other Works : Software

Josh Rehak, Weixiong Zheng, Alexander Blank, Ramirez de Chanlette, R. N. Slaybaugh. “BART.” Software (released 2019)

<https://github.com/SlaybaughLab/BART>

James Bevins, Youdong Zhang, and Rachel Slaybaugh. “Coeus.” Software. (released 2017) <https://github.com/SlaybaughLab/Coeus>

James Bevins, Youdong Zhang, and Rachel Slaybaugh. “Gnowee.” Software. (released 2017) <https://github.com/SlaybaughLab/Gnowee>

Ryan M. Bergmann, Kelly L. Rowland, Nikola Radnović, Rachel N. Slaybaugh, Jasmina L. Vujić. “WARP.” Software (released 2017) <https://github.com/SlaybaughLab/warp>

“PyNE: The Nuclear Engineering Toolkit.” Software. (periodic contributions 2014-2017)

<https://github.com/pyne/pyne>

Other Works : Funding Opportunity Announcements

Rachel Slaybaugh, Joel Fetter, Curt Nehrkorn, Geoffrey Short. “Generating Electricity Managed by Intelligent Nuclear Assets (GEMINA).” Funding Opportunity No. DE-FOA-002174. (released Oct 2, 2019)

<https://arpa-e-foa.energy.gov/#FoaId4f8d5ac3-267d-4849-ad63-80f858761a74>

Rachel Slaybaugh, Joel Fetter, Geoffrey Short. “Request for Information (RFI) on Intelligent Analytics, Algorithms, and Maintenance to Optimize Operations in Advanced Nuclear Reactors.” Request for Information (RFI) DE-FOA-0002115. (released March 26, 2019)

<https://arpa-e-foa.energy.gov/#FoaIdad3db639-ff92-447b-8b59-cde43defa55a>

Rachel Slaybaugh. “Leveraging Innovations Supporting nuclear Energy.” Funding Opportunity No. DE-FOA-0001953, Initial Announcement. (released Dec 20, 2018)

<https://arpa-e-foa.energy.gov/#FoaIde8647d89-1cac-4b58-8622-1b04de8958c4>

Rachel Slaybaugh, JC Zhao. “Modeling Enhance Innovations Trailblazing Nuclear Energy Reinvigoration (MEITNER).” Funding Opportunity No. DE-FOA-0001798. (released Oct 20, 2017)

<https://arpa-e-foa.energy.gov/Default.aspx?Archive=1#FoaId9688fafc-3b63-42af-9786-77d930987b4a>

Other Works : Other Writings

Suzanne Baker, Jessica Lovering, and Rachel Slaybaugh. “Our Progressive Policy Agenda for Advanced Nuclear Energy.” Policy Report for Good Energy Collective. August 2, 2020. <https://www.goodenergycollective.org/policy/progressive-policy-agenda-for-advanced-nuclear-energy>

Suzanne Baker, Jessica Lovering, Rachel Slaybaugh, and Denia Djokic. “A Policy Pathway for Nuclear Justice.” Policy Report for Good Energy Collective. April 27, 2021. <https://www.goodenergycollective.org/policy/a-policy-pathway-for-nuclear-justice>